

```
!pip install bitsandbytes  
!pip install --upgrade datasets fsspec
```

```

Collecting bitsandbytes
  Downloading bitsandbytes-0.46.1-py3-none-manylinux_2_24_x86_64.whl.metadata (10 kB)
Requirement already satisfied: torch<3,>=2.2 in /usr/local/lib/python3.11/dist-packages (from bitsandbytes) (2.6.0+cu124)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.11/dist-packages (from bitsandbytes) (2.0.2)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.2->bitsandbytes) (3.18.0)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.2->bitsandbytes) (4.13.0)
Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.2->bitsandbytes) (3.5)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.2->bitsandbytes) (3.1.6)
Requirement already satisfied: fsspec in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.2->bitsandbytes) (2025.3.2)
Collecting nvidia-cuda-nvrtc-cu12==12.4.127 (from torch<3,>=2.2->bitsandbytes)
  Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cuda-runtime-cu12==12.4.127 (from torch<3,>=2.2->bitsandbytes)
  Downloading nvidia_cuda_runtime_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cuda-cupti-cu12==12.4.127 (from torch<3,>=2.2->bitsandbytes)
  Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-cudnn-cu12==9.1.0.70 (from torch<3,>=2.2->bitsandbytes)
  Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-cublas-cu12==12.4.5.8 (from torch<3,>=2.2->bitsandbytes)
  Downloading nvidia_cublas_cu12-12.4.5.8-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cufft-cu12==11.2.1.3 (from torch<3,>=2.2->bitsandbytes)
  Downloading nvidia_cufft_cu12-11.2.1.3-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-curand-cu12==10.3.5.147 (from torch<3,>=2.2->bitsandbytes)
  Downloading nvidia_curand_cu12-10.3.5.147-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cusolver-cu12==11.6.1.9 (from torch<3,>=2.2->bitsandbytes)
  Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-cuspars-cu12==12.3.1.170 (from torch<3,>=2.2->bitsandbytes)
  Downloading nvidia_cuspars-cu12-12.3.1.170-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Requirement already satisfied: nvidia-cusparselt-cu12==0.6.2 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.2->bitsandbytes) (0.6.2)
Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.2->bitsandbytes) (2.21.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.2->bitsandbytes) (12.4.127)
Collecting nvidia-nvjitlink-cu12==12.4.127 (from torch<3,>=2.2->bitsandbytes)
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Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.11/dist-packages (from torch<3,>=2.2->bitsandbytes) (1.13.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy==1.13.1->torch<3,>=2.2->bitsandbytes) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages (from Jinja2->torch<3,>=2.2->bitsandbytes) (3.0.2)
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  72.9/72.9 MB 13.6 MB/s eta 0:00:00
Downloading nvidia_cublas_cu12-12.4.5.8-py3-none-manylinux2014_x86_64.whl (363.4 MB)
  363.4/363.4 MB 4.1 MB/s eta 0:00:00
Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (13.8 MB)
  13.8/13.8 MB 119.6 MB/s eta 0:00:00
Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (24.6 MB)
  24.6/24.6 MB 26.2 MB/s eta 0:00:00
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  883.7/883.7 kB 59.1 MB/s eta 0:00:00
Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl (664.8 MB)
  664.8/664.8 MB 907.2 kB/s eta 0:00:00
Downloading nvidia_cufft_cu12-11.2.1.3-py3-none-manylinux2014_x86_64.whl (211.5 MB)
  211.5/211.5 MB 6.4 MB/s eta 0:00:00
Downloading nvidia_curand_cu12-10.3.5.147-py3-none-manylinux2014_x86_64.whl (56.3 MB)
  56.3/56.3 MB 15.1 MB/s eta 0:00:00
Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-manylinux2014_x86_64.whl (127.9 MB)
  127.9/127.9 MB 8.8 MB/s eta 0:00:00
Downloading nvidia_cuspars-cu12-12.3.1.170-py3-none-manylinux2014_x86_64.whl (207.5 MB)
  207.5/207.5 MB 7.3 MB/s eta 0:00:00
Downloading nvidia_nvjitlink_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (21.1 MB)
  21.1/21.1 MB 105.8 MB/s eta 0:00:00
Installing collected packages: nvidia-nvjitlink-cu12, nvidia-curand-cu12, nvidia-cufft-cu12, nvidia-cuda-runtime-cu12, nvidia-cuda-cupti-cu12, nvidia-cudnn-cu12, nvidia-cublas-cu12, nvidia-cuspars-cu12
Successfully installed nvidia-cublas-cu12-12.4.5.8 nvidia-cuda-cupti-cu12-12.4.127 nvidia-cuda-nvrtc-cu12-12.4.127 nvidia-cuda-runtime-cu12-12.4.127 nvidia-cudnn-cu12-9.1.0.70 nvidia-cufft-cu12-11.2.1.3 nvidia-curand-cu12-10.3.5.147 nvidia-cusolver-cu12-11.6.1.9 nvidia-cuspars-cu12-12.3.1.170 nvidia-nvjitlink-cu12-12.4.127
Attempting uninstall: nvidia-nvjitlink-cu12
Found existing installation: nvidia-nvjitlink-cu12 12.5.82
Uninstalling nvidia-nvjitlink-cu12-12.5.82:
Successfully uninstalled nvidia-nvjitlink-cu12-12.5.82
Attempting uninstall: nvidia-curand-cu12
Found existing installation: nvidia-curand-cu12 10.3.6.82
Uninstalling nvidia-curand-cu12-10.3.6.82:
Successfully uninstalled nvidia-curand-cu12-10.3.6.82
Attempting uninstall: nvidia-cufft-cu12
Found existing installation: nvidia-cufft-cu12 11.2.3.61
Uninstalling nvidia-cufft-cu12-11.2.3.61:
Successfully uninstalled nvidia-cufft-cu12-11.2.3.61
Attempting uninstall: nvidia-cuda-runtime-cu12
Found existing installation: nvidia-cuda-runtime-cu12 12.5.82
Uninstalling nvidia-cuda-runtime-cu12-12.5.82:
Successfully uninstalled nvidia-cuda-runtime-cu12-12.5.82
Attempting uninstall: nvidia-cuda-nvrtc-cu12
Found existing installation: nvidia-cuda-nvrtc-cu12 12.5.82
Uninstalling nvidia-cuda-nvrtc-cu12-12.5.82:
Successfully uninstalled nvidia-cuda-nvrtc-cu12-12.5.82
Attempting uninstall: nvidia-cuda-cupti-cu12
Found existing installation: nvidia-cuda-cupti-cu12 12.5.82
Uninstalling nvidia-cuda-cupti-cu12-12.5.82:
Successfully uninstalled nvidia-cuda-cupti-cu12-12.5.82
Attempting uninstall: nvidia-cublas-cu12
Found existing installation: nvidia-cublas-cu12 12.5.3.2
Uninstalling nvidia-cublas-cu12-12.5.3.2:
Successfully uninstalled nvidia-cublas-cu12-12.5.3.2
Attempting uninstall: nvidia-cuspars-cu12
Found existing installation: nvidia-cuspars-cu12 12.5.1.3
Uninstalling nvidia-cuspars-cu12-12.5.1.3:

```

```

Successfully uninstalled nvidia-cusparse-cu12-12.5.1.3
Attempting uninstall: nvidia-cudnn-cu12
Found existing installation: nvidia-cudnn-cu12 9.3.0.75
Uninstalling nvidia-cudnn-cu12-9.3.0.75:
Successfully uninstalled nvidia-cudnn-cu12-9.3.0.75
Attempting uninstall: nvidia-cusolver-cu12
Found existing installation: nvidia-cusolver-cu12 11.6.3.83
Uninstalling nvidia-cusolver-cu12-11.6.3.83:
Successfully uninstalled nvidia-cusolver-cu12-11.6.3.83
Successfully installed bitsandbytes-0.46.1 nvidia-cublas-cu12-12.4.5.8 nvidia-cuda-cupti-cu12-12.4.127 nvidia-cuda-nvrtc-cu12-12.4.127
Requirement already satisfied: datasets in /usr/local/lib/python3.11/dist-packages (2.14.4)
Collecting datasets
  Downloading datasets-3.6.0-py3-none-any.whl.metadata (19 kB)
Requirement already satisfied: fsspec in /usr/local/lib/python3.11/dist-packages (2025.3.2)
Collecting fsspec
  Downloading fsspec-2025.5.1-py3-none-any.whl.metadata (11 kB)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from datasets) (3.18.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.11/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.11/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.3.7)
Requirement already satisfied: pandas in /usr/local/lib/python3.11/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.11/dist-packages (from datasets) (2.32.3)
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.11/dist-packages (from datasets) (4.67.1)
Requirement already satisfied: xxhash in /usr/local/lib/python3.11/dist-packages (from datasets) (3.5.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.70.15)
  Downloading fsspec-2025.3.0-py3-none-any.whl.metadata (11 kB)
Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.33.1)
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages (from datasets) (24.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-packages (from datasets) (6.0.2)
Requirement already satisfied: aiohttp!=4.0.0a0,!<4.0.0a1 in /usr/local/lib/python3.11/dist-packages (from fsspec[http]<=2025.3.0) (4.0.0a1)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.24.0) (4.12.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.2 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.24.0) (1.1.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2) (2025.11.11)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-packages (from pandas) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (from pandas) (2025.2)
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Requirement already satisfied: aiohappyeyeballs>=2.3.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!<4.0.0a1) (2.4.4)
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Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!<4.0.0a1) (25.3.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!<4.0.0a1) (1.4.1)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!<4.0.0a1) (6.1.0)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!<4.0.0a1) (0.2.0)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!<4.0.0a1) (1.18.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.8.2) (1.17.0)
  Downloading datasets-3.6.0-py3-none-any.whl (491 kB)
491.5/491.5 kB 37.2 MB/s eta 0:00:00
  Downloading fsspec-2025.3.0-py3-none-any.whl (193 kB)
193.6/193.6 kB 19.4 MB/s eta 0:00:00
Installing collected packages: fsspec, datasets
  Attempting uninstall: fsspec
    Found existing installation: fsspec 2025.3.2
    Uninstalling fsspec-2025.3.2:
      Successfully uninstalled fsspec-2025.3.2
  Attempting uninstall: datasets
    Found existing installation: datasets 2.14.4
    Uninstalling datasets-2.14.4:
      Successfully uninstalled datasets-2.14.4
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the
gcsfs 2025.3.2 requires fsspec==2025.3.2, but you have fsspec 2025.3.0 which is incompatible.
Successfully installed datasets-3.6.0 fsspec-2025.3.0

```

```

KeyboardInterrupt                                Traceback (most recent call last)
/tmp/ipython-input-5-1487276085.py in <cell line: 0>()
      1 get_ipython().system('pip install bitsandbytes')
----> 2 get_ipython().system('pip install --upgrade datasets fsspec')

```

```

10 frames
/usr/lib/python3.11/email/parser.py in parse(self, fp, headersonly)
    51     feedparser._set_headersonly()
    52     while True:
----> 53         data = fp.read(8192)
    54         if not data:
    55             break

```

KeyboardInterrupt:

```
import torch
from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForCausalLM, TrainingArguments, Trainer, BitsAndBytesConfig
import numpy as np
from peft import get_peft_model, LoraConfig, TaskType
```

```
model_name = 'TinyLlama/TinyLlama-1.1B-Chat-v1.0'
```

```
bnb_config = BitsAndBytesConfig(
    load_in_4bit = True,
    bnb_4bit_quant_type = 'nf4',
    bnb_4bit_compute_dtype = torch.bfloat16
)
```

```
model = AutoModelForCausalLM.from_pretrained(
    model_name,
    quantization_config = bnb_config,
    device_map = 'auto',
    trust_remote_code = True
)
```

```
tokenizer = AutoTokenizer.from_pretrained(model_name, trust_remote_code=True)
```

⚡ /usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as :
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(

```
lora_config = LoraConfig(
    r = 8,
    lora_alpha = 16,
    target_modules = ['q_proj', 'v_proj'],
    lora_dropout = 0.05,
    bias = 'none',
    task_type = TaskType.CAUSAL_LM
)
```

```
model = get_peft_model(model, lora_config)
```

```
data = load_dataset('openai/gsm8k', 'main', split='train[:200]')
```

⚡ README.md: 100% 7.94k/7.94k [00:00<00:00, 742kB/s]
train-00000-of-00001.parquet: 100% 2.31M/2.31M [00:00<00:00, 40.6MB/s]
test-00000-of-00001.parquet: 100% 419k/419k [00:00<00:00, 23.3MB/s]
Generating train split: 100% 7473/7473 [00:00<00:00, 9942.55 examples/s]
Generating test split: 100% 1319/1319 [00:00<00:00, 71257.46 examples/s]

```
def tokenize(batch):
    texts = [
        f"### Instruction:\n{inst}\n### Response:\n{out}"
        for inst, out in zip(batch['question'], batch['answer'])
    ]

    tokens = tokenizer(
        texts,
        padding = 'max_length',
        truncation = True,
        max_length = 256,
        return_tensors = 'pt'
    )

    tokens['labels'] = tokens['input_ids'].clone()

    return tokens
```

```
tokenized_data = data.map(tokenize, batched=True, remove_columns=data.column_names)
```

⚡ Map: 100% 200/200 [00:00<00:00, 1930.57 examples/s]

```

training_args = TrainingArguments(
    output_dir = './tinyllama-lora-tuned',
    per_device_train_batch_size = 4,
    gradient_accumulation_steps = 4,
    learning_rate = 1e-3,
    num_train_epochs = 50,
    fp16 = True,
    logging_steps = 20,
    save_strategy = 'epoch',
    report_to = 'none',
    remove_unused_columns = False,
    label_names = ["labels"]
)

```

```

trainer = Trainer(
    model = model,
    args = training_args,
    train_dataset = tokenized_data,
    processing_class = tokenizer
)

```

```
trainer.train()
```

 [600/600 17:54, Epoch 46/50]

Step	Training Loss
------	---------------

20	1.918600
40	0.813600
60	0.712200
80	0.625300
100	0.527800
120	0.453700
140	0.379200
160	0.296100
180	0.247900
200	0.188900
220	0.159300
240	0.123500
260	0.105800
280	0.083500
300	0.070800
320	0.058100
340	0.052300
360	0.045900
380	0.041600
400	0.037900
420	0.035900
440	0.034100
460	0.032900
480	0.031700
500	0.031000
520	0.030500
540	0.029600
560	0.029400
580	0.029000
600	0.028700

TrainOutput(global_step=600, training_loss=0.24182695905367532, metrics={'train_runtime': 1077.7647, 'train_samples_per_second': 2.278, 'train_steps_per_second': 0.557, 'total_flos': 1.4685733501071068e+16, 'train_loss': 0.24182695905367532, 'train_loss_std': 0.00010000000000000001})

```
model.save_pretrained("./tinyllama-lora-tuned-adapter-math")
tokenizer.save_pretrained("./tinyllama-lora-tuned-adapter-math")

↗ ('./tinyllama-lora-tuned-adapter-math/tokenizer_config.json',
 './tinyllama-lora-tuned-adapter-math/special_tokens_map.json',
 './tinyllama-lora-tuned-adapter-math/tokenizer.model',
 './tinyllama-lora-tuned-adapter-math/added_tokens.json',
 './tinyllama-lora-tuned-adapter-math/tokenizer.json')
```

✓ Evaluate

```
import os
import math

import torch
from torch.utils.data import DataLoader

from datasets import load_dataset
from transformers import AutoModelForCausalLM, AutoTokenizer, BitsAndBytesConfig, default_data_collator

from peft import PeftModel
```

```
os.getcwd()
```

```
↗ './content'
```

```
model_name = 'TinyLlama/TinyLlama-1.1B-Chat-v1.0'
adapter_path = './tinyllama-lora-tuned-adapter-math'
```

```
bnb_config = BitsAndBytesConfig(
    load_in_4bit = True,
    bnb_4bit_quant_type = 'nf4',
    bnb_4bit_compute_dtype = torch.bfloat16
)
```

```
tokenizer = AutoTokenizer.from_pretrained(model_name, trust_remote_code=True)
```

```
↗ /usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as :
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
```

```
warnings.warn(
```

```
tokenizer_config.json:      1.29k/? [00:00<00:00, 121kB/s]
```

```
tokenizer.model: 100%                               500k/500k [00:00<00:00, 24.9MB/s]
```

```
tokenizer.json:      1.84M/? [00:00<00:00, 46.3MB/s]
```

```
special_tokens_map.json: 100%                       551/551 [00:00<00:00, 55.4kB/s]
```

```
base_model = AutoModelForCausalLM.from_pretrained(
    model_name,
    quantization_config = bnb_config,
    device_map = 'auto',
    trust_remote_code = True
).eval()
```

```
↗ config.json: 100%                                608/608 [00:00<00:00, 52.0kB/s]
```

```
model.safetensors: 100%                            2.20G/2.20G [00:12<00:00, 44.8MB/s]
```

```
generation_config.json: 100%                      124/124 [00:00<00:00, 2.59kB/s]
```

```
tmp_model = AutoModelForCausalLM.from_pretrained(
    model_name,
    quantization_config = bnb_config,
    device_map = 'auto',
    trust_remote_code = True
)
```

```
tuned_model = PeftModel.from_pretrained(tmp_model, adapter_path)
```

```
tuned_model = tuned_model.merge_and_unload().eval()
```

```
→ /usr/local/lib/python3.11/dist-packages/peft/tuners/lora/bnb.py:351: UserWarning: Merge lora module to 4-bit linear may get differer
warnings.warn(
```

```
def tokenize(batch):
    texts = [
        f"### Instruction:\n{inst}\n### Response:\n{out}"
        for inst, out in zip(batch['question'], batch['answer'])
    ]
```

```
    tokens = tokenizer(
        texts,
        padding = 'max_length',
        truncation = True,
        max_length = 256,
        return_tensors = 'pt'
    )
```

```
    tokens['labels'] = tokens['input_ids'].clone()
```

```
    return tokens
```

```
eval_ds = load_dataset('openai/gsm8k', 'main', split='train[:20]')
eval_ds = eval_ds.map(tokenize, batched=True, remove_columns=['question', 'answer'])
eval_ds = eval_ds.with_format('torch')
```

```
→ README.md:      7.94k/? [00:00<00:00, 441kB/s]

main/train-00000-of-00001.parquet: 100%                2.31M/2.31M [00:01<00:00, 30.6kB/s]
main/test-00000-of-00001.parquet: 100%                 419k/419k [00:01<00:00, 323kB/s]
Generating train split: 100%                          7473/7473 [00:00<00:00, 4834.48 examples/s]
Generating test split: 100%                            1319/1319 [00:00<00:00, 20958.41 examples/s]

Map: 100%                                             20/20 [00:00<00:00, 138.81 examples/s]
```

```
eval_loader = DataLoader(
    eval_ds,
    batch_size = 8,
    collate_fn = default_data_collator
)
```

```
@torch.no_grad()
def compute_perplexity(model):
    losses = []

    for batch in eval_loader:
        batch = {k: v.to('cuda') for k, v in batch.items()}
        loss = model(**batch).loss
        losses.append(loss.item())
```

```
    return math.exp(sum(losses) / len(losses))
```

```
print(f'Base Model Perplexity: {compute_perplexity(base_model):.2f}')
print(f'Tuned Model Perplexity: {compute_perplexity(tuned_model):.2f}')
```

```
→ Base Model Perplexity: 139.67
  Tuned Model Perplexity: 1.05
```

```
import random
```

```
raw_data = load_dataset('gsm8k', 'main', split='train[:20]')
refs = raw_data['answer']
```

```
def generate(model, instruction):
    token_ids = tokenizer(f"### Instruction:\n{instruction}\n### Response:\n", return_tensors='pt').input_ids.to('cuda')

    with torch.no_grad():
        out = model.generate(token_ids, max_new_tokens=256)
```